

JavaScript Practice Problems

PROBLEM 01 ■ EASY - Reading List

```
let readingList = {  
  books: [],  
  totalBooks: 0  
};
```

Write:

- addBook(title, author) — adds a book object, updates totalBooks
- showList() — prints each book with its number (1, 2, 3...)

Edges:

- Empty list
- Same book added twice (allow it)

PROBLEM 02 ■ EASY - Simple Scoreboard

```
let scoreboard = {  
  players: [],  
  roundsPlayed: 0  
};
```

Write:

- addPlayer(name)
- addPoints(name, points)
- showScoreboard()

Edges:

- Player not found
- Negative points
- Empty scoreboard

PROBLEM 03 ■■ MEDIUM - Grocery List Manager

```
let groceries = {  
  items: [],  
  totalItems: 0,  
  purchasedCount: 0  
};
```

Write:

- addItem(name, quantity)
- markPurchased(name)
- showList()
- showProgress()

Edges:

- Item doesn't exist
- Already purchased
- Empty list

PROBLEM 04 ■■ MEDIUM - Student Grade Tracker

```
let gradeBook = {  
  students: [],  
  classAverage: 0  
};
```

Write:

- addStudent(name, score)
- updateClassAverage()
- showTopStudent()
- showGradeBook()

Edges:

- No students
- Equal top score
- Score above 100

PROBLEM 05 ■■ MEDIUM - Task Manager

```
let taskManager = {  
  tasks: [],  
  completedCount: 0,  
  pendingCount: 0  
};
```

Write:

- addTask(title, priority)
- completeTask(title)
- showByPriority(priority)
- showSummary()

Edges:

- Task not found
- Already completed
- No priority matches

PROBLEM 06 ■■ MEDIUM - Expense Logger

```
let expenseLog = {  
  entries: [],  
  totalSpent: 0,  
  categoryBreakdown: {}  
};
```

Write:

- addExpense(category, description, amount)
- showByCategory(category)
- showTotal()

Edges:

- Empty category
- Amount 0
- Negative amount

PROBLEM 07 ■■ MEDIUM - Contact Book

```
let contactBook = {  
  contacts: [],
```

```
totalContacts: 0
};
```

Write:

- addContact(name, phone, city)
- findContact(name)
- showByCity(city)
- deleteContact(name)

Edges:

- Contact not found
- Delete non-existent contact
- City with no contacts

PROBLEM 08 ■■■ CHALLENGE - Movie Recommendation Engine

```
let movieDb = {
  movies: [],
  watchedCount: 0
};
```

Write:

- addMovie(title, genre, rating)
- markWatched(title)
- recommend(genre)
- showStats()

Edges:

- No unwatched movies
- Genre not found
- Already watched
- Empty database

Hint:

Use manual sorting by rating.

PROBLEM 09 ■■■ CHALLENGE - Full Habit Tracker

```
let habitTracker = {
  habits: [],
  totalHabits: 0,
  todayCompleted: 0
};
```

Build:

- addHabit()
- markDone()
- resetDay()
- showHabits()
- showStats()

Edges:

- Habit not found
- Already completed
- Reset empty
- No habits

PROBLEM 10 ■■■ CHALLENGE - Mini Budget Tracker

```
let budget = {  
  transactions: [],  
  balance: 0,  
  totalIncome: 0,  
  totalExpenses: 0  
};
```

Build:

- addIncome()
- addExpense()
- showTransactions()
- showSummary()

Edges:

- Expense > balance
- Income 0
- No transactions
- Only income